

Linux Class 6 - Task

Users, Groups, Ownership and Permissions

For practice I made a folder called class6 with a file called file1.txt inside it. I also made a group called developers and three practice users named arjun, priya and rahul. After finishing the task I deleted the three users and the group again, so my system is back to normal.

One thing about this whole task: adding users, removing users and changing the owner of a file all need sudo, because only the administrator is allowed to do them. It asks for my password the first time and then remembers it for a few minutes. chmod is different, it does not need sudo as long as I own the file.

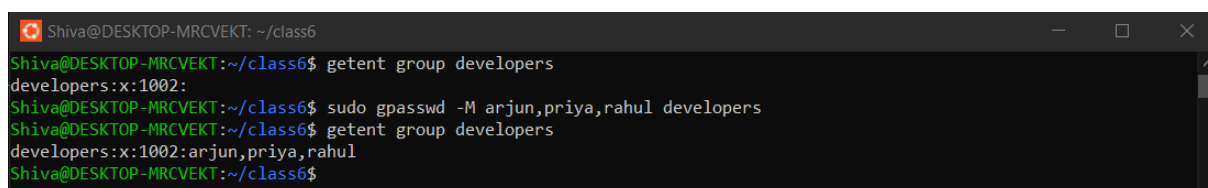
1. How to add multiple users together to a group?

I used the gpasswd command with the -M option:

```
sudo gpasswd -M arjun,priya,rahul developers
```

- gpasswd is the command that handles who is in a group.
- -M means set the member list.
- The user names are separated by commas with no spaces in between.
- The group name comes last.

Before running it, getent group developers showed nothing after the last colon, which means the group was empty. After running it, the same command showed all three names.

A terminal window titled 'Shiva@DESKTOP-MRCVEKT: ~/class6'. The first command is 'getent group developers', which outputs 'developers:x:1002:'. The second command is 'sudo gpasswd -M arjun,priya,rahul developers'. The third command is 'getent group developers', which now outputs 'developers:x:1002:arjun,priya,rahul'. The prompt returns to 'Shiva@DESKTOP-MRCVEKT:~/class6\$'.

Screenshot 1: The group was empty, and after one command it has all three users.

One thing I had to be careful about: -M replaces the whole list. If somebody was already in the group and I do not write their name, they are removed. If I only want to add one more person without disturbing the others, I use this instead:

```
sudo usermod -aG developers rahul
```

Here -a means append, which is add. If I forget the -a and write only -G, the user is removed from all his other groups, which is a mistake that is easy to make.

2. How to remove multiple users together from a group?

To remove one user the option is -d, which means delete:

```
sudo gpasswd -d rahul developers
```

There is no option to give many names at once for removing, so for more than one user I ran the same command inside a small loop:

```
for u in arjun priya; do sudo gpasswd -d $u developers; done
```

The loop takes each name from the list one by one, puts it in place of \$u, and runs the command. Each time it printed a line saying the user was removed. After that the group was empty again.

```
Shiva@DESKTOP-MRCVEKT: ~/class6
Shiva@DESKTOP-MRCVEKT:~/class6$ sudo gpasswd -d rahul developers
Removing user rahul from group developers
Shiva@DESKTOP-MRCVEKT:~/class6$ getent group developers
developers:x:1002:arjun,priya
Shiva@DESKTOP-MRCVEKT:~/class6$ for u in arjun priya; do sudo gpasswd -d $u developers; done
Removing user arjun from group developers
Removing user priya from group developers
Shiva@DESKTOP-MRCVEKT:~/class6$ getent group developers
developers:x:1002:
Shiva@DESKTOP-MRCVEKT:~/class6$
```

Screenshot 2: One user removed with -d, then two more removed with a loop.

Another way to empty a group completely in one command is:

```
sudo gpasswd -M "" developers
```

The two quotation marks with nothing inside mean an empty member list.

3. What will these commands do?

a) sudo chown john:developers file1.txt

It changes the owner of file1.txt to the user john, and at the same time changes the group of the file to developers. The part before the colon is the user and the part after the colon is the group. It needs sudo because giving a file to another user is an administrator job. I ran the same command with arjun instead of john, because john was not a user on my system.

b) chmod o-rwx file1.txt

It takes away read, write and execute from others. Others means everybody who is not the owner and is not in the group of the file. My file started as -rw-r--r-- and became -rw-r-----, so the last three letters became three dashes.

c) chmod u+x,g+x file1.txt

It adds the execute permission to the owner and to the group. The comma lets me write two changes in one command. My file became -rwxr-x---. The same thing can be written shorter as chmod ug+x file1.txt.

```
Shiva@DESKTOP-MRCVEKT: ~/class6
Shiva@DESKTOP-MRCVEKT:~/class6$ ls -l file1.txt
-rw-r--r-- 1 Shiva Shiva 0 Sep 28 16:05 file1.txt
Shiva@DESKTOP-MRCVEKT:~/class6$ chmod o-rwx file1.txt
Shiva@DESKTOP-MRCVEKT:~/class6$ ls -l file1.txt
-rw-r----- 1 Shiva Shiva 0 Sep 28 16:05 file1.txt
Shiva@DESKTOP-MRCVEKT:~/class6$ chmod u+x,g+x file1.txt
Shiva@DESKTOP-MRCVEKT:~/class6$ ls -l file1.txt
-rwxr-x--- 1 Shiva Shiva 0 Sep 28 16:05 file1.txt
Shiva@DESKTOP-MRCVEKT:~/class6$
```

Screenshot 3: The permissions before, after o-rwx, and after u+x,g+x.

These are the letters used with chmod:

Letter	What it means
u	The user who owns the file.
g	The group the file belongs to.
o	Others, meaning everybody else.
a	All three of them together.
+	Add this permission.
-	Take this permission away.

Letter	What it means
=	Set it to exactly this and nothing else.

4. How to change the owner of a file?

The command is chown, which is short for change owner. If I write only a user name, only the owner changes:

```
sudo chown priya file1.txt
```

In my screenshot the owner changed from arjun to priya but the group stayed developers, because I did not mention any group.

```

Shiva@DESKTOP-MRCVEKT: ~/class6
Shiva@DESKTOP-MRCVEKT:~/class6$ ls -l file1.txt
-rwxr-x--- 1 Shiva Shiva 0 Sep 28 16:05 file1.txt
Shiva@DESKTOP-MRCVEKT:~/class6$ sudo chown arjun:developers file1.txt
Shiva@DESKTOP-MRCVEKT:~/class6$ ls -l file1.txt
-rwxr-x--- 1 arjun developers 0 Sep 28 16:05 file1.txt
Shiva@DESKTOP-MRCVEKT:~/class6$ sudo chown priya file1.txt
Shiva@DESKTOP-MRCVEKT:~/class6$ ls -l file1.txt
-rwxr-x--- 1 priya developers 0 Sep 28 16:05 file1.txt
Shiva@DESKTOP-MRCVEKT:~/class6$

```

Screenshot 4: chown with a user and a group, and then chown with only a user.

The different forms are:

- sudo chown priya file1.txt changes the owner only.
- sudo chown priya:developers file1.txt changes the owner and the group.
- sudo chown :developers file1.txt changes only the group.
- sudo chgrp developers file1.txt also changes only the group.

I can always check the result with ls -l file1.txt. The first name it shows is the owner and the second name is the group.

Commands I used

Command	What it does
sudo gpasswd -M a,b,c grp	Sets the members of the group to a, b and c.
sudo usermod -aG grp user	Adds one user without removing the others.
sudo gpasswd -d user grp	Removes one user from the group.
sudo gpasswd -M "" grp	Empties the group completely.
getent group grp	Shows who is in the group.
groups user	Shows which groups a user belongs to.
sudo chown user file	Changes the owner only.
sudo chown user:grp file	Changes the owner and the group.
sudo chown :grp file	Changes only the group.
chmod o-rwx file	Takes all permissions away from others.
chmod u+x,g+x file	Gives execute to the owner and the group.
ls -l file	Shows the owner, the group and the permissions.